

Influence of Torque and Cadence on Power Output Production in Cyclists

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Purpose: No information is available on the torque/cadence relationship in road cyclists. We aimed to establish whether this relationship differs between cyclists of different performance levels or team roles. **Methods:** Mean maximal power (MMP) output data from 177 riders were obtained from 2012 to 2021 from training and competitions. Cyclists were categorized according to their performance level (world-tour [WT, $n = 68$], procontinental [PC, $n = 63$], or under 23 [U23, $n = 46$]) and team role (time trialists [$n = 12$], all-rounders [$n = 94$], climbers [$n = 64$], or team leaders [$n = 7$]). **Results:** A significant interaction effect was found for absolute and relative MMP ($P < .001$), with higher values in PC than WT for short (5–60 s) efforts and the opposite trend for longer durations. MMP was also greater in PC than in U23 for short efforts (30–60 s), with WT and PC attaining higher MMP than U23 for longer bouts (5–60 min). A significant interaction effect was found for cadence ($P = .007$, but with no post hoc differences) and absolute ($P = .010$) and relative torque ($P = .002$), with PC and WT showing significantly higher torque (all $P < .05$) than U23 for 5- to 60-minute efforts, yet with no differences between the former 2 performance levels. No interaction effect between team roles was found for cadence ($P = .185$) or relative torque ($P = .559$), but a significant interaction effect was found for absolute torque ($P < .001$), with all-rounders attaining significantly higher values than climbers for 5-second to 5-minute efforts. **Conclusions:** Differences in MMP between cycling performance levels and rider types are dependent on torque rather than cadence, which might support the role of torque development in performance.

Keywords: performance, professional cycling, mean maximal power output

Measuring power output (PO) under field conditions (ie, during racing and training) with mobile power meters provides new opportunities to investigate the performance requirements of cycling.^{1–3} Indeed, PO has proven to be an important metric for determining race demands and for characterizing rider types and cycling performance levels.^{4,5} PO can be simplified as the product of torque and pedaling rate (cadence), but it is not known which of these 2-factor conditions the observed differences between cyclists of different performance levels. Hence, analyzing the relationship between torque and cadence for a given effort duration in elite road cyclists could provide important insights in this respect.

In cycling, torque is the rotational force applied to the pedals, which is averaged to 1-pedal revolution.⁶ Few cycle manufacturers provide torque values as a data output source due to intellectual property policies and commercial interests. Consequently, the majority of available research on the torque and cadence relationship has been conducted in the laboratory using force-velocity,⁷ isokinetic,⁸ or inertial load⁹ sprinting-type (< 6 s) efforts. An exception to this is a study by Gardner et al,¹⁰ who analyzed both 6-second laboratory and 65-m field-sprinting efforts to compare the torque and cadence relationship under the 2 conditions. Bergin-Taylor et al¹¹ recently investigated the torque-cadence relationship in professional cyclists using different 6-second sprint modalities (seated vs standing and high vs low cadence). Most of the previously mentioned research has focused only on sprint-type efforts, and to our knowledge no study has analyzed the torque and cadence relationship for prolonged effort durations (5 min–1 h). Such information might aid in understanding the decline in PO after accumulated work,^{2,5,12,13} and could be used as a training tool to inform on cyclists' strengths and weaknesses with respect to PO production capabilities.^{14,15}

The aim of the present study was to establish whether the torque and cadence relationship for different effort durations differs between cyclists of different performance levels (ie, international cycling federation-licensed world tour [WT], procontinental [PC],

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and under 23 [U23] level cyclists), or between team roles (ie, all-rounders, time trialists, climbers, and team leaders). Based on the existing literature on the observed differences in the PO profile and endurance durability both between cyclists of different performance levels and between different rider types,^{2,12,13,16,17} we hypothesized that the torque and cadence relationship would also be significantly different, particularly for the longest efforts.

Methods

Study Design and Participants

The PO data from 177 cyclists were obtained over 9 years (2012–2021), resulting in a total of 161,744 analyzed individual PO files (885 [762] per cyclist), which were obtained from both training sessions and competitions. The cyclists were members of 5 cycling teams of 3 different performance levels (WT, *n* = 68; PC, *n* = 63; U23, *n* = 46). As explained by Valenzuela et al,⁴ participants were classified attending to their team roles as either *all-rounders* (*n* = 94, cyclists who do not especially excel in high mountains or time trials, but can achieve good results in mixed stages [of whom 29, 37, and 28 were classified as WT, PC, and U23]), *time trialists* (*n* = 12, who are required to perform maximally in time trials [of whom 7, 3, and 2 were classified as WT, PC, and U23]), *climbers* (*n* = 64, who are required to perform maximally in high mountain ascents [of whom 25, 23, and 16 were classified as WT, PC, and U23]), or *team leaders* (*n* = 7, capable of winning a Grand Tour [of whom 7, 0, and 0 were classified as WT, PC, and U23]). The protocol complied with the Declaration of Helsinki, and all participants provided written informed consent, and the local Ethics Committee approved the protocol.

Measures

Participants' height was assessed at the beginning of the study, and body weight was assessed once a week (and entered in the corresponding power meter accordingly) using a standard stadiometer and scale, respectively. PO was registered during all possible training sessions and races using 5 different brands of power meter (Power2-Max Type SNieder; SRAM Red, Quarq; Shimano Dura-Ace FCRC9100-P, Shimano; SRM; and Pioneer SGY-PM910H2, Pioneer). All power meters have previously been used in studies reported in the scientific literature,^{18,19} and all were factory calibrated at least once per season. A zero-offset was performed before each session in accordance with the manufacturers' instructions. PO data

were checked for potential spikes using specific software (WKO5 Build 576, TrainingPeaks LLC) and manually corrected when necessary. When erroneous outliers were recorded, data were manually corrected using the Data Spike ID and FIX chart. If the number of outliers was greater than 5% of the total data to be analyzed, the value was eliminated, and the data were considered missing.

We registered the highest PO values (mean maximal power [MMP], expressed both in absolute terms and relative to cyclists' body mass) obtained for efforts lasting between 5 seconds and 60 minutes. Efforts of longer durations were removed from analyses to ensure that we only recorded maximal efforts. For each effort duration, we computed torque (expressed both in absolute units and relative to cyclists' body mass) and cadence (in rpm) from PO values using the following equation: Torque = PO ÷ (cadence × $\pi \div 30$), as explained by Gardner et al.¹⁰ All cyclists used a standard crank length of 170 to 175 mm.

Statistical Analysis

Data are shown as mean (SD). Normality was tested using the Kolmogorov–Smirnov test, and homoscedasticity was assessed using Levene test. Differences between cyclists' performance levels and team roles for descriptive characteristics (eg, age and weight) were assessed with a 1-way analysis of variance (ANOVA). Differences between cyclists' performance levels and team roles for MMP, cadence, and torque were assessed using 2-way mixed ANOVA, with effort duration as the within-subject factor and performance level or team role as the between-subject factor. To minimize the risk of statistical type I errors, post hoc comparisons were only performed (using the Bonferroni test) when a significant main group (ie, performance level or team role, for the 1-way ANOVA) or group (performance level or team role) by effort duration interaction effect (for the 2-way mixed ANOVA) was found. Statistical analyses were conducted using SPSS (version 23.0) statistical software setting the significance level at 0.05.

Results

Differences Between Performance Levels

No significant differences between performance levels were found for height, weight, or body mass index. WT cyclists were significantly older than PC and U23 peers, with significant differences also found between the latter groups (Table 1).

Table 1 Descriptive Characteristics of Study Participants (N = 177) According to Their Performance Level and Team Role

Performance level	Age, y	Weight, kg	Height, cm	Body mass index, kg/m ²
WT (<i>n</i> = 68)	32.5 (5.4) ^{#†}	66.7 (6.0)	179 (7)	20.8 (1.2)
PC (<i>n</i> = 63)	25.9 (3.8) ^{*†}	66.0 (6.0)	177 (6)	20.9 (1.3)
U23 (<i>n</i> = 46)	20.1 (1.5) [*]	67.2 (7.2)	180 (6)	19.8 (4.5)
Team role				
All-rounders (<i>n</i> = 94)	25.8 (6.0) [§]	69.6 (5.5) ^{‡§}	181 (6) ^{‡§}	21.3 (1.3) [‡]
Time trialists (<i>n</i> = 12)	29.4 (7.5)	70.9 (6.4) ^{‡§}	182 (6) ^{‡§}	21.5 (1.0) [‡]
Climbers (<i>n</i> = 64)	27.5 (6.5)	61.8 (4.0) ^{§¶}	176 (5) ^{§¶}	19.4 (3.7) ^{§¶}
Team leaders (<i>n</i> = 7)	32.7 (5.2)	62.0 (2.5)	174 (5)	20.5 (0.7)

Abbreviations: PC, procontinental; U23, under 23; WT, world tour. Data are shown as mean (SD).

Significant differences (all *P* < .05): *Differences with WT. #Differences with PC. †Differences with U23. ‡Differences with climbers. §Differences with team leaders.

¶Differences with all-rounders. ¶Differences with time trialists.

Table 2 Differences in MMP Values and Cadence Between the Different Performance Levels

Effort duration	WT			PC			U23		
	MMP, W	MMP, W/kg	Cadence, rpm	MMP, W	MMP, W/kg	Cadence, rpm	MMP, W	MMP, W/kg	Cadence, rpm
5 s	1173 (159)	17.6 (1.7)#	102 (12)	1220 (165)	18.5 (2.0)	105 (9)	1238 (171)	18.4 (1.6)	104 (9)
30 s	808 (109)#	12.1 (1.3)#	91 (7)	866 (110)†	13.1 (1.2)†	93 (8)	815 (104)	12.2 (1.3)	94 (7)
1 min	650 (82)#	9.8 (1.2)#	88 (6)	692 (95)	10.5 (1.1)†	87 (7)	658 (71)	9.8 (1.1)	89 (7)
5 min	474 (334)†	7.1 (0.5)†	86 (7)	469 (46)†	7.1 (0.5)†	85 (8)	447 (34)	6.7 (0.4)	86 (7)
10 min	439 (30)†	6.6 (0.5)†	87 (6)	427 (38)†	6.5 (0.5)†	85 (7)	404 (38)	6.0 (0.5)	87 (7)
20 min	407 (30)†	6.1 (0.5)†	88 (6)	397 (37)†	6.0 (0.5)†	85 (8)	376 (32)	5.6 (0.4)	85 (7)
30 min	386 (30)†	5.8 (0.5)†	88 (6)	372 (37)†	5.7 (0.5)†	86 (7)	352 (36)	5.3 (0.4)	86 (8)
1 h	352 (30)#†	5.3 (0.5)#†	88 (6)#	334 (26)†	5.1 (0.3)†	85 (6)	318 (35)	4.7 (0.4)	85 (7)

Abbreviations: MMP, mean maximal power; PC, procontinental; U23, under 23; WT, world tour. Data are shown as mean (SD).
 Significant differences (all $P < .05$): *Significant differences with WT. #Significant differences with PC. †Significant differences with U23.

A significant interaction effect between performance levels was found for MMP expressed both in absolute ($P < .001$) and relative ($P = .001$) values (Table 2). Absolute and relative MMP achieved was significantly higher for PC cyclists than for WT peers for short-duration efforts (5 s–1 min). The opposite trend was observed with increasing effort durations, with both performance levels achieving similar absolute and relative MMP for efforts lasting 5 to 30 minutes, and with WT cyclists achieving significantly higher MMP than PC cyclists for 60-minute efforts. MMP achieved was also significantly higher for PC cyclists than for U23 cyclists for short-duration efforts (30 s and 1 min), with both professional groups attaining higher MMP than U23 cyclists for longer efforts (5–60 min).

Analysis of the PO components showed a significant interaction effect for cadence ($P = .007$), although *post hoc* analyses revealed a significantly higher cadence in WT cyclists than in PC cyclists only for 60-minute efforts (Table 2). A significant interaction effect was also found for torque expressed both in absolute ($P = .010$) and

relative ($P = .002$) units (Figures 1 and 2, respectively). Overall, PC cyclists attained the highest absolute and relative torque values for short-duration efforts, with significantly higher torque values than both U23 and WT cyclists for 30-second and 1-minute efforts. Moreover, both PC and WT cyclists attained significantly higher absolute and relative torque values than U23 cyclists for longer efforts (most efforts ranging between 5 and 60 min), with no differences between the former 2 performance levels.

Differences Between Team Roles

Significant differences between team roles were found for all descriptive variables. Notably, time trialists and all-rounders were the heaviest and tallest cyclists (with a significantly higher weight and height than both climbers and team leaders), and both team roles had a significantly higher body mass index than climbers (Table 1).

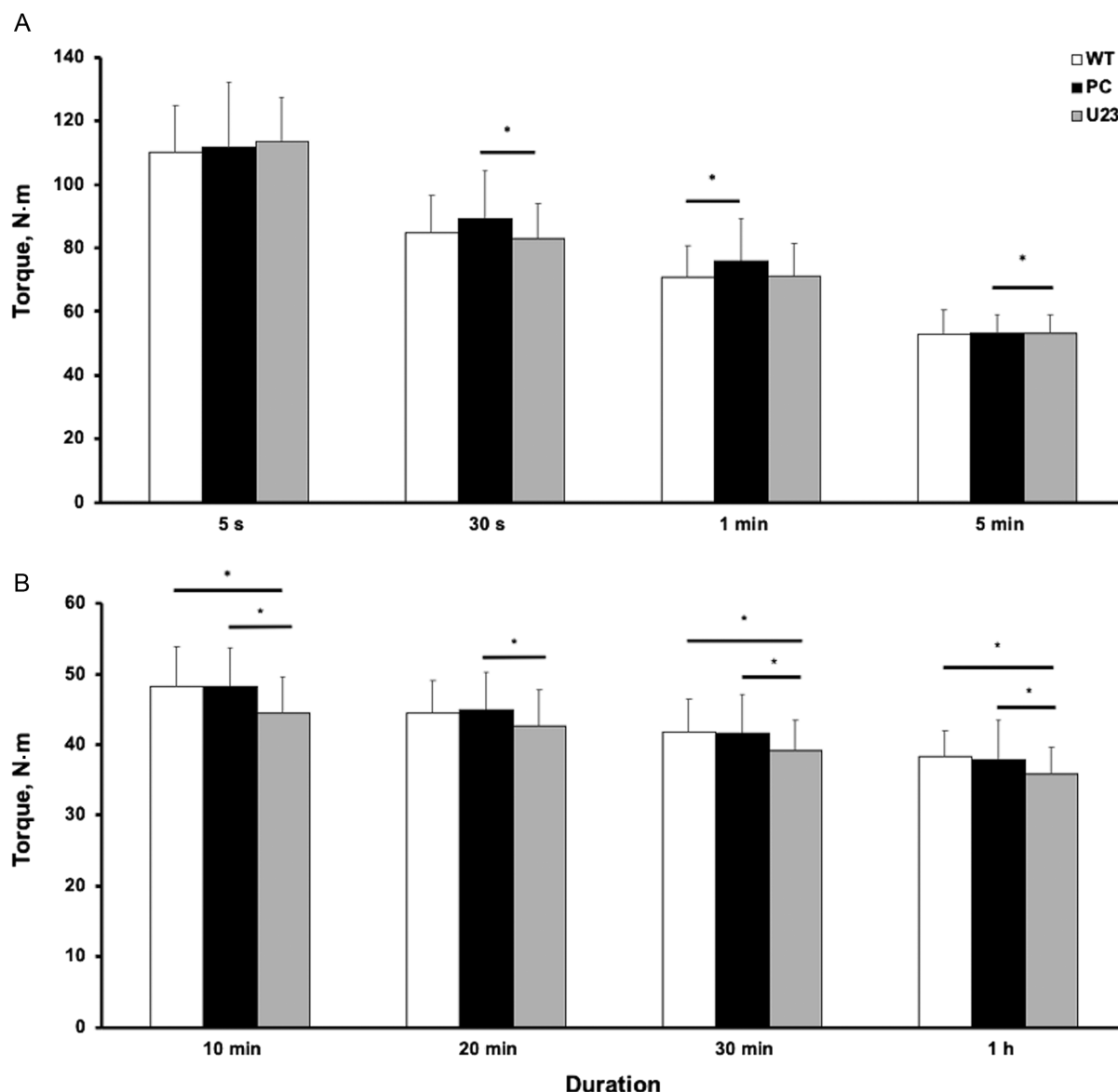


Figure 1 — Differences in absolute torque between the different cycling performance levels. PC indicates procontinental; U23, under 23; WT, world tour. *Significant differences (all $P < .05$).

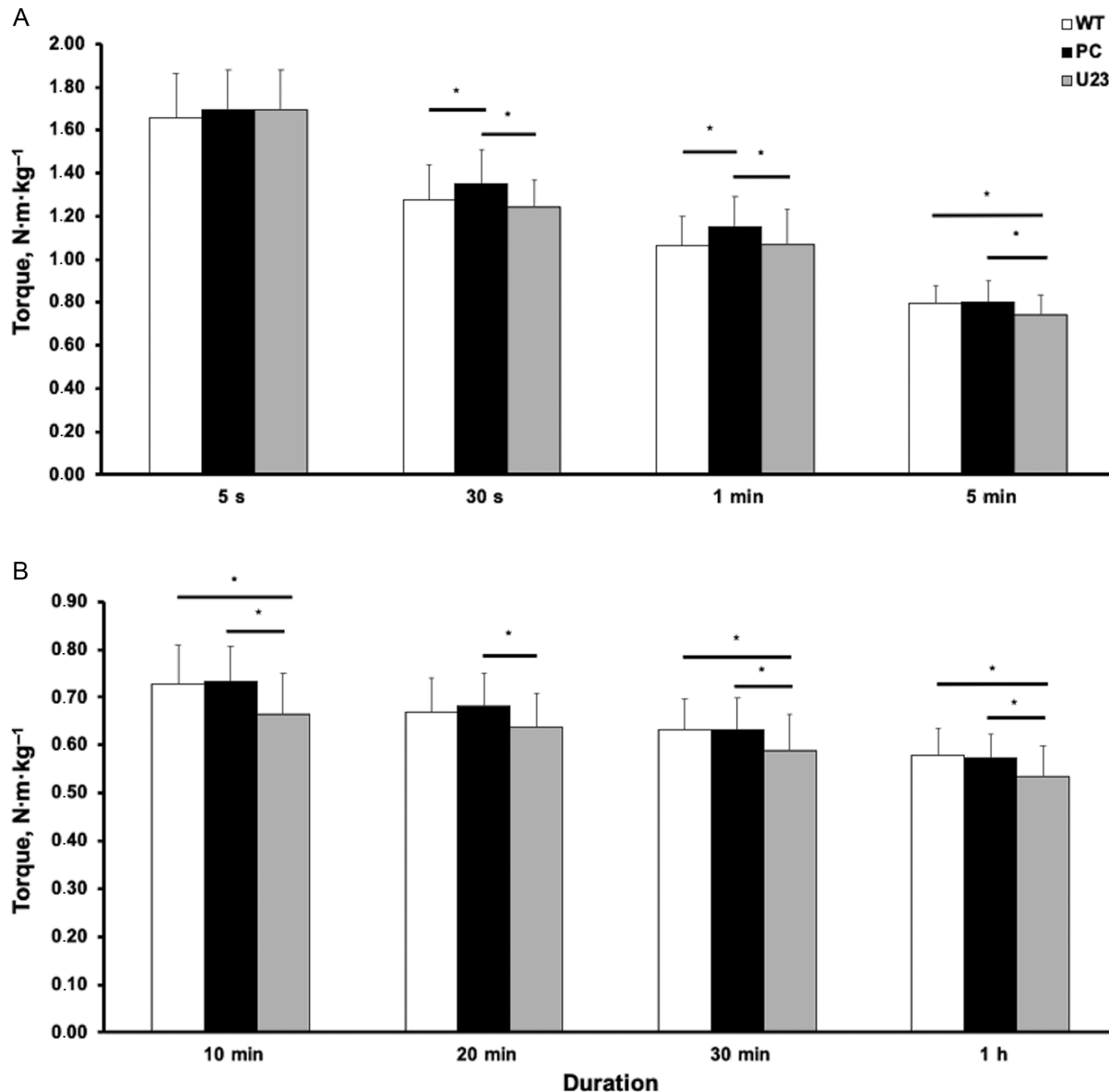


Figure 2 — Differences in relative torque between the different cycling performance levels. PC indicates procontinental; U23, under 23; WT, world tour. *Significant differences (all $P < .05$).

A significant interaction effect between team roles was found for MMP expressed both in absolute ($P < .001$) and relative ($P = .001$) units (Table 3). All-rounders attained significantly higher absolute MMP than climbers for all effort durations, as well as higher absolute MMP than team leaders for the shortest effort duration (5 s). Time trialists also attained higher absolute MMP than climbers for the longest efforts (60 min). The opposite trend was observed for relative MMP, as both climbers and, particularly, team leaders attained higher relative MMP than both all-rounders and time trialists for efforts lasting between 5 and 60 minutes.

No significant interaction effect between team roles was found for cadence ($P = .185$; Table 2) or torque expressed in relative units ($P = .559$; Figure 3), but a significant interaction effect was found for torque expressed in absolute units ($P < .001$; Figure 4). Post hoc analyses revealed that all-rounders attained significantly higher absolute torque than climbers for efforts lasting between 5 seconds and 5 minutes, with no more differences between team roles.

Discussion

Our findings reveal differences in MMP between cyclists based on performance levels or team roles, which can be explained by differences in absolute and relative torque but not in cadence.

Riders' Performance Level or Team Role

Our findings of higher absolute and relative MMP in WT and PC cyclists compared with U23 ones are in line with previous studies.^{2,13,20} In turn, absolute and relative MMP differed between WT and PC depending on effort duration, with longer efforts favoring the latter. Overall, these findings might highlight that differences in the absolute and relative power profile between PC and WT groups might be explained, at least partly, by different team roles and therefore event-specific specialization, with a previous study by Sanders et al^{3,21} indeed reporting a differing power profile between

Table 3 Differences in MMP Values and Cadence Between the Different Team Roles

Effort duration	All-rounders			Time trialists			Climbers			Team leaders		
	MMP, W	MMP, W/kg	Cadence, rpm	MMP, W	MMP, W/kg	Cadence, rpm	MMP, W	MMP, W/kg	Cadence, rpm	MMP, W	MMP, W/kg	Cadence, rpm
5 s	1285 (155)‡§	18.5 (1.9)	105 (10)	1212 (120)	17.1 (1.5)	103 (14)	1103 (133)	17.9 (1.7)	101 (9)	1104 (88)	17.8 (1.2)	103 (8)
30 s	874 (105)‡	12.6 (1.4)	94 (8)	818 (85)	11.6 (1.0)	95 (10)	775 (98)	12.5 (1.2)	91 (7)	775 (83)	12.5 (1.1)	87 (10)
1 min	698 (79)‡	10.1 (1.2)	89 (6)	669 (89)	9.5 (1.0)	89 (6)	623 (79)	10.1 (1.1)	87 (7)	645 (48)	10.4 (0.7)	87 (6)
5 min	477 (40)‡	6.8 (0.4)‡§	85 (8)	471 (42)	6.7 (0.6)‡§	86 (7)	447 (36)	7.3 (0.5)	87 (6)	457 (21)	7.4 (0.3)	81 (6)
10 min	434 (37)‡	6.2 (0.5)‡§	87 (7)	437 (38)	6.2 (0.7)‡§	89 (7)	412 (36)	6.7 (0.5)	86 (7)	426 (19)	6.9 (0.3)	83 (6)
20 min	402 (36)‡	5.8 (0.5)‡§	86 (7)	411 (42)	5.8 (0.7)§	90 (7)	383 (31)	6.2 (0.4)\$	86 (7)	398 (13)	6.4 (0.2)	85 (5)
30 min	378 (37)‡	5.4 (0.5)‡§	87 (7)	383 (43)	5.4 (0.8)§	89 (7)	361 (33)	5.8 (0.5)\$	87 (6)	376 (14)	6.1 (0.3)	86 (6)
1 hr	342 (31)‡	4.9 (0.4)‡§	87 (7)	354 (48)‡	5.0 (0.8)§	88 (7)	326 (32)	5.3 (0.4)\$	85 (6)	346 (8)	5.6 (0.2)	87 (5)

Abbreviation: MMP, mean maximal power. Note: Data are shown as mean (SD).
Significant differences (all $P < .05$): ‡Differences with climbers. §Differences with team leaders. \$Differences with all-rounders.

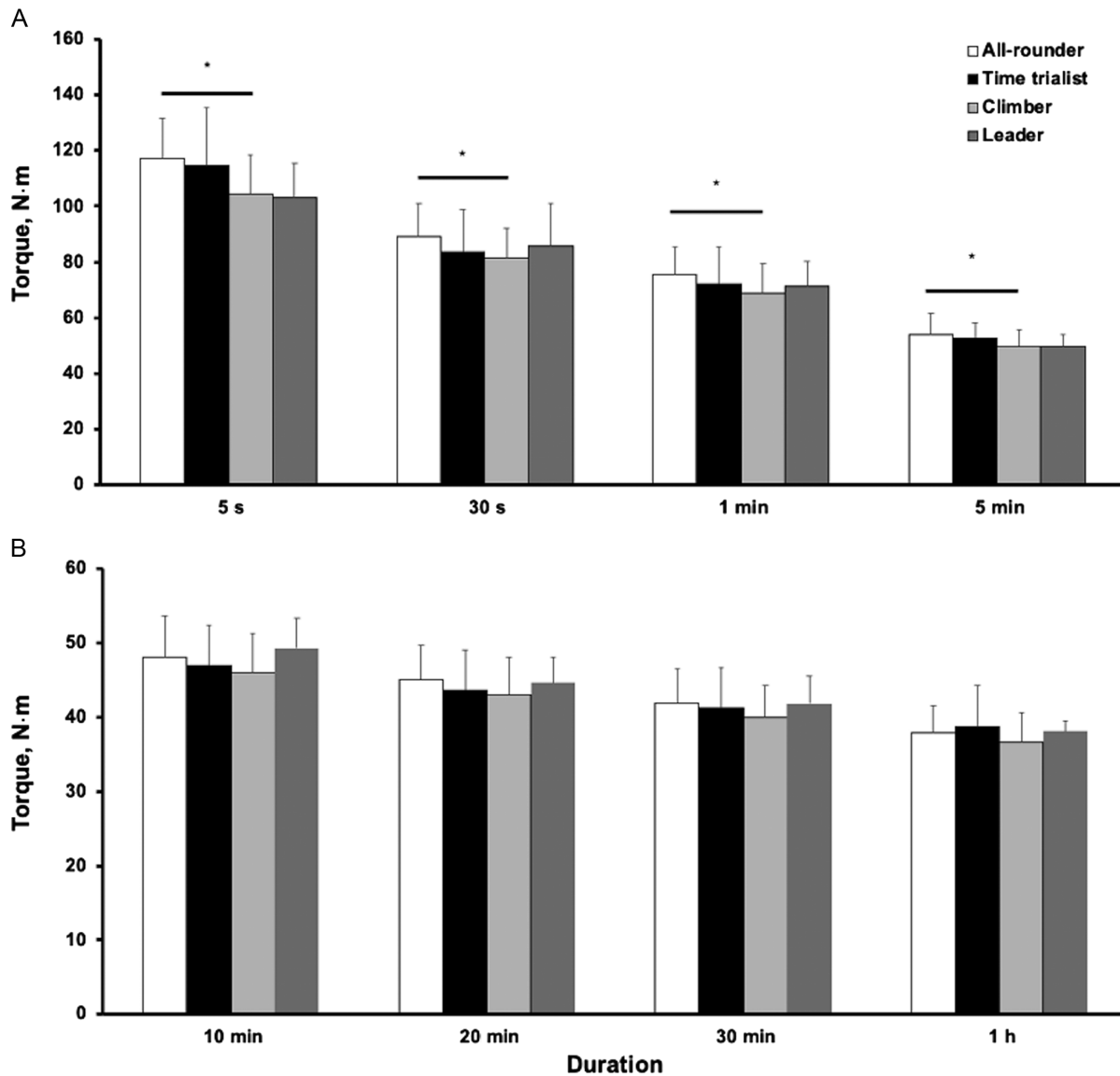


Figure 3 — Differences in absolute torque between the different team roles. *Significant differences between all-rounders and climbers (all $P < .05$).

stage types and performance level categories. On the other hand, improvements in the absolute or relative power profile only occur on the torque production capability and not on the cadence side proportion. Indeed, PC and WT cyclists attained significantly higher absolute and relative torque values than U23 cyclists for most efforts ranging between 5 and 60 minutes, with no differences between the former 2 performance levels, and with no differences in cadence between the 3 performance levels. In this regard, further improvements in absolute torque are difficult to achieve once an elite cycling level has been reached, so that optimization in body composition is needed to increase relative torque values. In this effect, Leo et al recently reported no significant differences in absolute MMP values (for efforts lasting 2–12 min) throughout the season in U23 cyclists whereas, due to decreases in cyclists' body weight during the season, relative 12-minute MMP was higher in late season compared with early season.²²

For all effort durations, all-rounders and time trialists recorded the highest absolute MMP profile, whereas team leaders had the highest relative MMP profile. These findings also align with

previous research,^{3,4,12,23,24} which found differences in the MMP between not only different cycling performance levels, but also team roles, or typologies. Although previous research reported differences in cadence profiles between flat and uphill stages and between different ascent categories (first vs second category),^{17,25,26} cadence profiles in our study did not differ based on riders' performance level or team role. As PO is a product of pedaling velocity (cadence) and force (torque), the differences accounted in PO can only be explained by differences in torque production. For this reason, understanding the PO production capability based on the torque and cadence relationship might also help to identify cyclists' strengths and weaknesses. Improvements in endurance "durability" might only be achieved through increasing the torque rather than cadence.

Torque and Cadence Time Relationship

To the best of our knowledge, this is the first study investigating the torque and cadence relationship in elite-/international-level

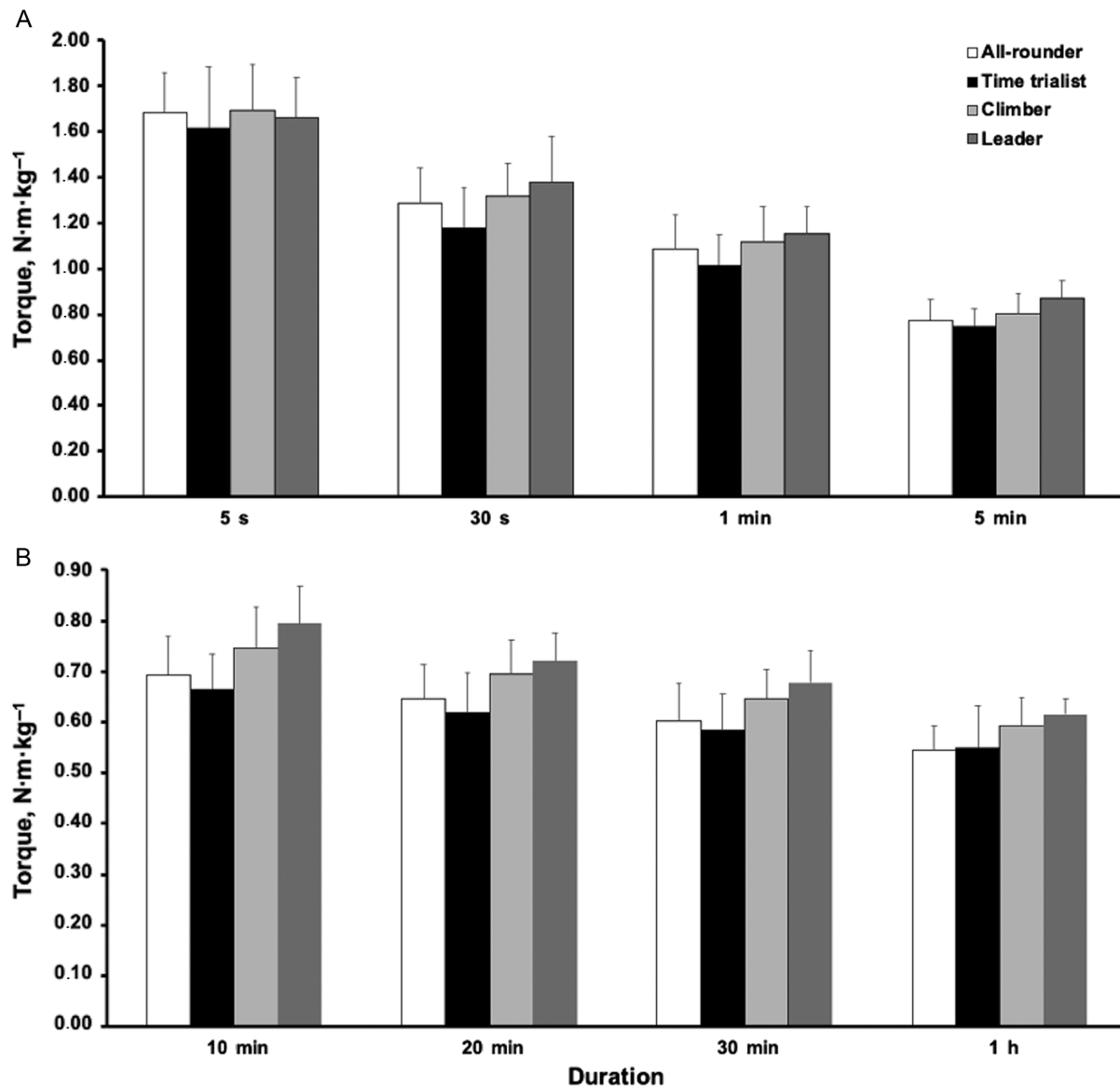


Figure 4 — Differences in relative torque between the different team roles. No significant differences were found.

road cyclists. Gardner et al¹⁰ reported that maximum torque values derived from laboratory and field tests can be used interchangeably for sprint-type efforts (< 6 s); however, most of the torque and cadence relationship research has been examined under laboratory conditions or in track cyclists.²⁷ Indeed, previous research^{28–30} has frequently used laboratory-based isokinetic tests (eg, knee extension) for measuring maximum torque, to determine peripheral or central fatigue mechanisms for different endurance sports (eg, cycling and running). Similar to the phase transition in the critical power concept between steady-state and nonsteady state exercise,^{31,32} the same physiological responses could be found with critical torque during maximum isokinetic knee extensor exercise.²⁹ These findings imply that maintaining torque production capability is directly linked to whole-body exercise perturbations. Several recent studies have reported a decline in the MMP profile after accumulated work and intensity for professional road cyclist,^{2,12–15} and so analyzing the torque and cadence relationship in conjunction with the

MMP profiles might help to better understand the underlying mechanisms of endurance “durability”—that is, the ability to attenuate the deterioration in physiological (eg, increased heart rate) or performance (eg, reduced PO) indicators under fatigue conditions.³³

Limitations

We are aware of the descriptive nature of the study, which can be considered a study limitation. Also, analyzing the torque and cadence relationship through controlled laboratory tests might have yielded more comprehensive insights on this topic. In turn, the large sample size analyzed and the high level of the included cyclists—including professional teams from multiple countries—should be considered as a major strength. Moreover, despite the limitations of analyzing field-based PO data, we set strict methodological procedures (eg, calibration of power meters and data filtering) to ensure high data quality.

Practical Applications

The present findings can help practitioners to better determine PO production capabilities based on the torque–time and cadence–time relationship for different rider types and performance levels (U23, PT, and WT). This information might be beneficial to improve durability by placing emphasis on the maintenance of torque production capabilities for short (1 s–5 min) and long (5–60 min) duration efforts. Although more research is needed in this regard, increases in torque production capabilities might be achieved by including strength training strategies (eg, off-bike resistance training or on-bike isometric cycling)^{34–36} or biomechanical adjustments (eg, adopting crank length and pedal speed).^{6,35,36} However, given that scientific evidence for these strategies is currently not strong enough to provide clear recommendations, future research should investigate how different strategies can improve torque production capabilities.

Conclusion

The findings of the present study indicate that differences in MMP between performance levels and team roles are solely dependent on torque values rather than on variations in cadence. For this reason, emphasis should be placed on torque production development, as the best riders seem able to attain higher torque values on long-duration efforts.

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